
Deep Learning 2026: Multimodal Forecasting and Classification of Glaucomatous Visual Fields

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1 Introduction and Background

Problem statement. Glaucoma is a degenerative optic neuropathy characterized by progressive functional vision loss. It is a leading cause of irreversible blindness, requiring early detection to prevent vision loss. Monitoring relies on Visual Field (VF) perimetry. Given a patient’s current functional and temporal data, we formalize three predictive tasks:

1. **Classification:** Map current VF $x_{hvf} \in \mathbb{R}^{8 \times 9}$ to severity categories $y_{cls} \in \{\text{Mild, Moderate, Severe}\}$.
2. **Multimodal Forecasting:** Predict future Mean Total Deviation $y_{mtd} \in \mathbb{R}$ and future VF grid $y_{hvf} \in \mathbb{R}^{8 \times 9}$ from patient VF, patient age, years since first measurement, and years since last measurement: $x_{i,t} = \{H_{i,t}, m_{i,t}\}$ where $H_{i,t} \in \mathbb{R}^{8 \times 9}$ and $m_{i,t} = [\text{age}_i, \Delta t_{first}, \Delta t_{last}]$.
3. **Forecasting from optic disc images:** Generate VF representations $y_{vf} \in \mathbb{R}^{61 \times 61}$ conditioned on high-resolution optic disc images $x_{image} \in \mathbb{R}^{224 \times 224}$.

Related work. DL algorithms achieve excellent performance in diagnosing glaucoma using fundus photography and OCT imaging [2]. But recent research in glaucoma has shifted from simple diagnostic classification—categorizing eyes as healthy or glaucomatous—to the more complex task of structure-function mapping and longitudinal forecasting. We implement a dual-pathway approach: first, translating structural anatomy into function by predicting current visual field (VF) grids from optic disc images; and second, forecasting future VF states by integrating baseline sensitivities with clinical metadata such as patient age and the time interval between visits.

2 Method

2.1 Model Architecture

Tasks 1 and 2: Our current unimodal baseline extracts spatial features from an 8×9 VF grid using a 3-block CNN (channels: 16, 32, 32) with ReLU activations. The flattened spatial embedding is concatenated with temporal scalar features: patient age, years since the first measurement, and years since the last measurement.

This unified feature vector $z \in \mathbb{R}^{2307}$ is passed through three independent heads: 1. **Classifier:** Linear layer outputting logits for 3 severity classes. 2. **MTD Regressor:** Linear layer predicting a continuous scalar. 3. **VF Forecaster:** MLP ($256 \rightarrow 72$ neurons) reshaped to output a $1 \times 8 \times 9$ spatial grid.

Task 3: We use an EfficientNet model as the backbone [4] and replace the classifier head by a linear layer, followed by ReLU and Dropout, and one final linear layer to output a 61×61 spatial grid.

2.2 Training Setup

Dataset UWHVF: For multimodal forecasting we use the University of Washington Humphrey Visual Field (UWHVF) [3] dataset. It includes 28943 VFs from 7248 eyes of 3871 patients. Progression was calculated for 2985 eyes from 1579 patients.

Dataset GRAPE: For predicting visual fields from optic discs, we use a public longitudinal glaucoma real-world appraisal progression ensemble (GRAPE) [1] dataset. The GRAPE dataset contains 1115 follow-up records from 263 eyes, with VFs, fundus images, OCT measurements and clinical information, and OD segmentation and VF progression are annotated.

Preprocessing: Visual fields, age, time since first/last measurement were all normalized. Optic disc images were normalized and resized to (224, 224). For forecasting tasks, the data was split by patient to avoid the model learning individual patient data rather than the pattern of the disease. 54-point Humphrey VFs were mapped to an 8×9 spatial grid, with non-tested anatomical blind spots padded with a default scalar of 100.

Loss function: For MTD and VF prediction, MSE loss was used in both datasets.

For classification, due to the imbalanced UWHVF data set (13,015 Mild; 4,722 Moderate; 3,778 Severe), the standard Cross-Entropy did not learn the boundaries of the Moderate class. We replaced it with a weighted Focal Loss:

$$\mathcal{L}_{focal} = -\frac{1}{N} \sum_{i=1}^N \alpha_{y_i} (1 - p_{t,i})^\gamma \log(p_{t,i}) \quad (1)$$

where $\alpha = [1.0, 2.76, 3.44]$ acts as the inverse class frequency weight, and $\gamma = 2.0$ down-weights easily classified examples. The total network loss is a weighted sum of the classification, MTD regression, and VF reconstruction losses:

$$\mathcal{L}_{total} = \mathcal{L}_{focal} + 0.5 \cdot \text{MSE}(y_{mtd}, \hat{y}_{mtd}) + 2.0 \cdot \text{MSE}(y_{hvf}, \hat{y}_{hvf}) \quad (2)$$

Optimizer and hyperparameters: The multi-task CNN model was trained using the PyTorch Lightning framework. To ensure reproducibility, below is a table detailing the training parameters.

Table 1 details the specific hyperparameters used to achieve the final reported results.

Table 1: Hyperparameters for the Multi-task CNN and EfficientNet

Parameter	Multi-task CNN	EfficientNet
Optimizer	Adam	Adam
Learning Rate	0.001	0.001
Batch Size	32	32
Max Epochs	70	100

2.3 Implementation Details

We tried adding gaussian noise or random affine transformations to augment the data, but this yielded a degradation in the performance of the models.

3 Results

3.1 Main Results

Our multi-task CNN achieves moderate to high accuracy on current VF classification and low Mean Squared Error (MSE) on future VF forecasting. Evaluation on the unseen test sets is presented in the table below.

Table 2: Multi-task CNN test results on UWHVF and EfficientNet results on GRAPE

Metric	Multi-task CNN	EfficientNet
Classification Accuracy		
Overall	84.11%	-
Mild (Class 0)	85.77%	-
Moderate (Class 1)	72.39%	-
Severe (Class 2)	79.70%	-
Forecasting Error (MSE)		
Mean Total Deviation (MTD)	0.0068	-
Visual Field Grid (HVF)	0.0112	0.0133

3.2 Analysis

Classification The model achieved an accuracy of 84.11% overall on the imbalanced dataset.

Multimodal forecasting The model predicts the MTD and VF better on healthy cases. It generally performs well for MTD prediction, but struggles to identify where the outlier values will be in glaucomatous VFs. This is expected as the latter is a difficult task. See difference VFs and MTD for all test inputs on GitHub.

Forecasting from optic disc images The model is able to learn to predict visual fields from optic disc images in some cases, as seen in the image below. However, on some inputs, the model just predicts the mean visual field and fails to detect glaucoma, see 2.

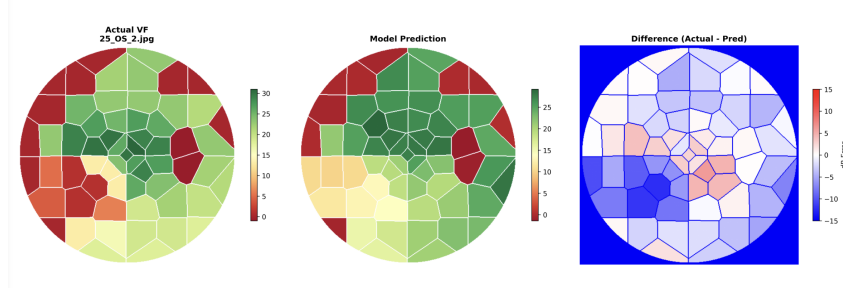


Figure 1: Actual vs model predicted VF, and difference plot — detection

4 Conclusion

We implemented a Multi-task CNN to classify visual fields into the categories mild, moderate, or severe. The same model was also trained to forecast MTD and VF progression. Additionally, we fine-tuned an EfficientNet model to predict a VF from an optic disc image.

- **Key findings:** Forecasting full VFs is a relatively hard task compared to predicting MTD or classifying VFs. It is also quite likely that structural data (optic disc images) do not have enough information to fully predict a well-defined visual field.
- **Limitations:** As is often the case in medical imaging, the amount of available data is rather low. Larger datasets would likely significantly improve model performance.
- **Possible improvements:** We briefly tried to fine-tune Vision Transformers (GenViT and HybViT) for task 3 but they ended up always predicting the mean value for all inputs. Further investigations into using Vision Transformers for that task could be done.

References

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A Dataset Preprocessing: HVF 24-2 to 2D Grid Representation

Standard Automated Perimetry (SAP) using the Humphrey Field Analyzer (HFA) 24-2 protocol tests 54 specific retinal locations. Because Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) expect dense, rectangular spatial grids, these 54 points must be embedded into an 8×9 matrix.

As implemented in our data pipeline, the 54 points are mapped to specific $(row, column)$ indices corresponding to their anatomical locations. The 8×9 grid contains 72 total cells. The 18 non-tested locations (representing the natural blind spot and the corners of the visual field) are padded with a default scalar value of 100.0. During the forward pass, the network learns to ignore these padded regions. Target values representing Mean Total Deviation (MTD) are mapped to discrete severity categories based on standard clinical thresholds:

- **Mild:** $MTD > -6.0$ dB
- **Moderate:** $-12.0 \text{ dB} \leq MTD \leq -6.0$ dB
- **Severe:** $MTD < -12.0$ dB

B Extended Results: Optic disc to VF

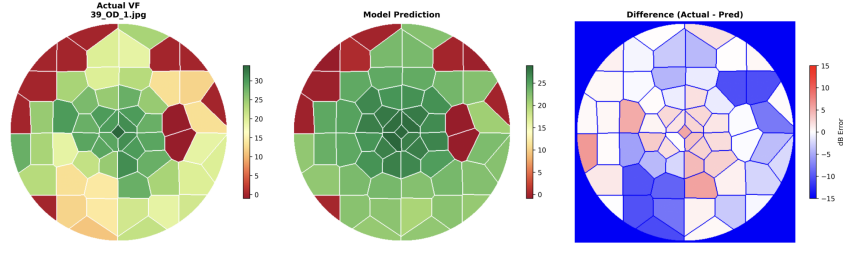


Figure 2: Actual vs model predicted VF, and difference plot — no detection

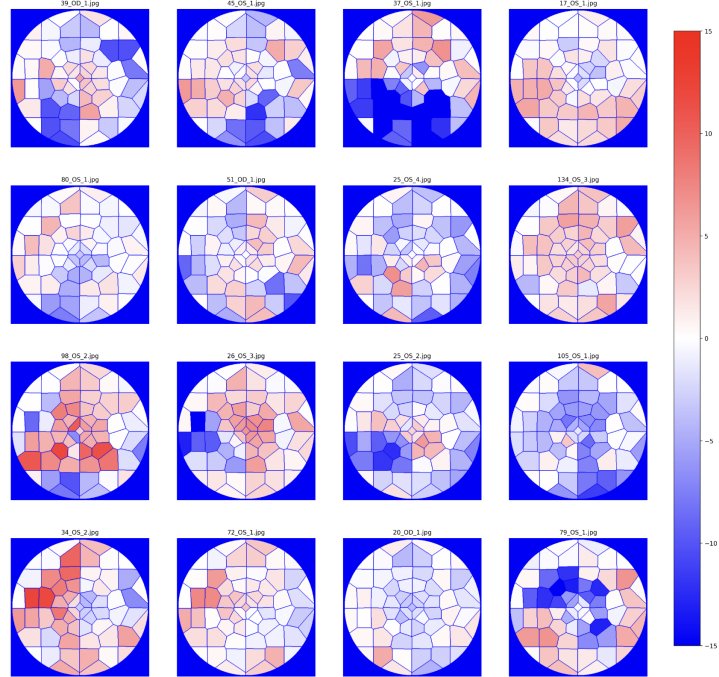


Figure 3: Difference plot for a batch of predictions

C Extended Results: Comparing Approaches

Fixing Ordinal Boundary Ambiguity. Initial experiments utilizing standard Cross-Entropy loss resulted in a severe accuracy paradox. Because the Mild class dominates the dataset, the model optimized for global accuracy by over-predicting Mild cases. Consequently, the Moderate class accuracy was originally 58.21%. By introducing the class-weighted Focal Loss, the model was heavily penalized for misclassifying the rarer boundary cases. This adjustment increased Moderate class accuracy to 72.39% (a +14.18% improvement) while preserving the overall global accuracy and maintaining stable regression metrics. We had also tested deriving classes from regression-based predictions to try and improve accuracy furthermore, but the regression-derived approach underperformed on Severe because the MD regression predicts the next visit’s MTD, while the true label is the current visit’s severity. So the denormalization thresholds are being applied to a future prediction, not the current state. Further research is to be made to increase the model’s performance.

To explicitly demonstrate the impact of the class-weighted Focal Loss on ordinal boundary ambiguity, Table 3 compares the per-class test accuracies of the model trained with standard unweighted Cross-Entropy (CE) versus the Regression-derived approach versus the proposed Focal Loss.

Table 3: Ablation study of Loss Functions on Test Set Classification Accuracy

Loss Function	Global Acc.	Mild (0)	Moderate (1)	Severe (2)
Standard Cross-Entropy	84.54%	91.47%	58.21%	80.18%
Regression-derived	82.41%	86.02%	71.79%	69.58%
Weighted Focal Loss (Ours)	84.11%	85.77%	72.39%	79.70%

While the standard CE loss achieved a marginally higher global accuracy (84.54%), this metric was artificially inflated by over-predicting the majority class (Mild), leaving the model effectively blind to the boundaries of the Moderate class (58.21%). The Focal Loss sacrificed a small fraction of Mild accuracy to achieve a substantial +14.18% gain in Moderate accuracy, resulting in a much more clinically reliable and balanced predictive distribution.